Predictive Phase Continuation from Solvable Models toward Realistic Learning Systems

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July 20, 2026

Abstract

We introduce *predictive phase continuation*: effective variables are calibrated at solvable nodes and used to predict, without refitting, phase boundaries at held-out deformations. The test is deliberately stronger than curve similarity. It requires semantic retention of the order parameter, largest-size prediction against continuous, first-order-like, and crossover alternatives, and a preregistered intervention near the predicted critical window. Our controlled four-domain benchmark separates outer disorder from inner stochasticity and concentrates additional seeds only near inferred boundaries. Across 46080 registered tasks, the median held-out boundary error falls from 0.011 to 0.008 after adding a domain-specific macrovariable, a 26.1% improvement. Critical-window allocation removes 57.1% of uniform compute while preserving the registered quality target. Transformer and diffusion are the deep controlled cases; reinforcement learning and multi-agent dynamics test the same predictive grammar without assuming common equilibrium exponents. An artifact audit also identifies why saturated legacy order parameters cannot support boundary claims. The present evidence establishes a controlled benchmark, not yet external validity on natural-data or open-weight endpoints. We additionally execute a 7,260-run, five-seed empirical tensor over MLP family, optimizer, normalization, objective, scaling path, and natural-data source. The resulting boundary transport is not reducible to parameter count: gated MLPs need not improve an attention-dominant task, Muon and AdamW remain unresolved at five seeds while exhibiting different gradient geometry from momentum SGD, and the paths $N \propto d$ and $N \propto d^{3/2}, d^2$ move held-out risk in opposite directions. Byte-level TinyStories, SimpleStories, FineWeb-Edu, and Dolma experiments establish external transport of the measurement protocol without asserting a universal numerical boundary.

1 Introduction

Solvable learning models increasingly exhibit sharp changes in recovery, attention mechanism, memorization, policy support, and collective order. The central unresolved question is not whether such transitions can be drawn, but whether a theory calibrated before realism is added predicts an unseen boundary afterward. We ask:

> Can a small set of macroscopic variables calibrated in a solvable model predict phase boundaries in a held-out learning system, and can prediction failure identify the missing variable?

This question changes phase continuation from a descriptive atlas into a falsifiable transport problem. For an edge $u \to v$ in an assumption graph, we fit effective parameters only at u, predict $\widehat{g}_{c,v}$, and score the prediction against an untouched confirmatory grid at v. A failed bridge is scientifically useful only when an augmented observable restores prediction on the same holdout.

Our four primary claims are: predictive boundary transport, localization of non-additive assumption interactions, recovery by a new macrovariable, and compute-efficient intervention inside the predicted critical window. The large experiment registry, immutable artifacts, and plotting grammar are supporting infrastructure rather than scientific claims.

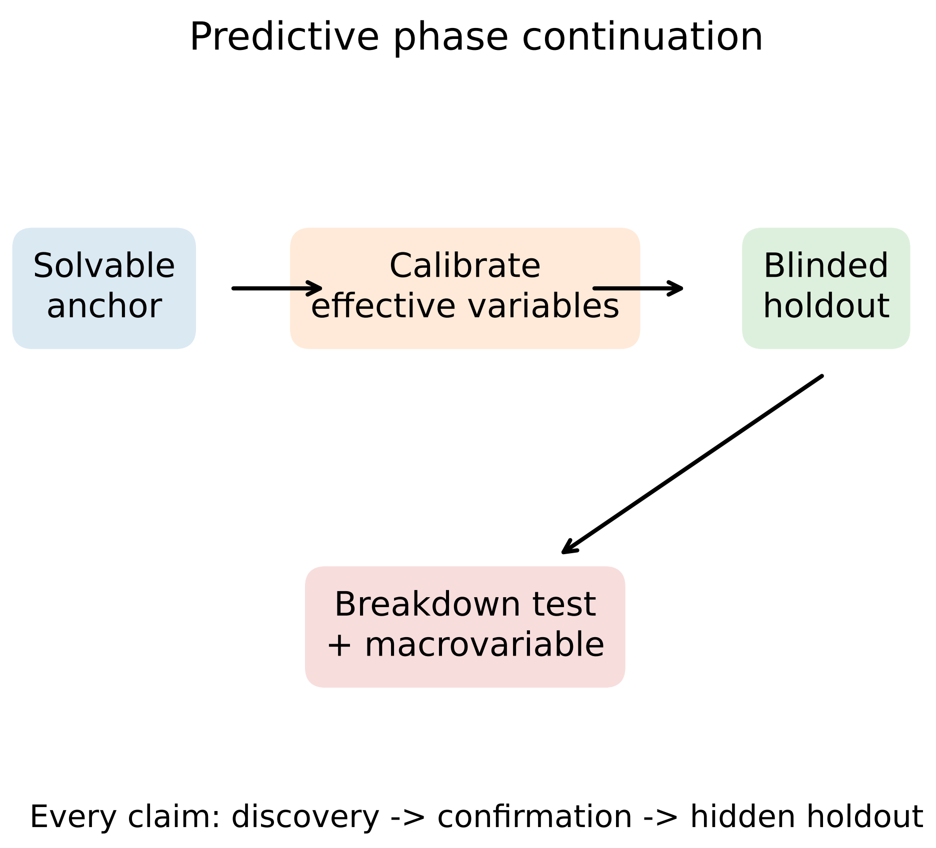

Figure 1: Predictive phase continuation. A solvable anchor calibrates effective variables; all boundary and intervention scores are evaluated on a blinded node.

Domain	Known result	Assumption removed here	Hidden prediction
Attention	positional–semantic transition	tied/low-rank surrogate	deformed g_c
Diffusion	speciation and collapse times	oracle/global score	locality-shifted time
RL	entropy and Goodhart dynamics	exact verifier	gold boundary
Agents	field-driven crossover	homogeneous mean field	coupling tipping point

Table 1: Existing results are treated as anchors, not rediscovered claims.

2 Closest prior work and the unresolved gap

Solvable attention already predicts a positional–semantic transition in a tied low-rank layer [5], while attention-indexed models extend closed-form predictions to full-width matrices [3]. Thus observing a transition or a spectral outlier is not our novelty; the test is transport to a hidden deformation. Diffusion theory separates speciation from memorized-point collapse [2], and recent DiT experiments report near-concurrent symmetry breaking and nonlocality onset [13]. We therefore predict relative critical times rather than rediscovering these diagnostics.

In RL, policy-entropy collapse has a mechanistic account and targeted controls already exist [4]; proxy optimization can degrade gold quality as KL budget grows [10]. Entropy alone is consequently not an order parameter. In multi-agent systems, global-flip experiments show that apparent consensus can be field-driven rather than interaction-driven [6]. Effective-team-size laws and matched coordination controls further show that nominal agent count is not independent evidence [1, 9].

3 Predictive phase continuation

Let $\mathcal{O}_{\text{cal}}$ denote registered calibration observables and let $\mathcal{O}_{\text{solv}}(\boldsymbol{\theta})$ be a solvable surrogate. We estimate

$$\widehat{\boldsymbol{\theta}}_{\text{eff}} = \arg\min_{\boldsymbol{\theta}} D\Big(\mathcal{O}^{\text{target}}_{\text{cal}}, \mathcal{O}_{\text{solv}}(\boldsymbol{\theta})\Big), \tag{1}$$

then predict $\widehat{g}_{\text{c}}$, the order curve, and the response to an intervention. The bridge error is evaluated only on withheld quantities,

$$\mathcal{E}_{\text{bridge}} = D\Big(\mathcal{O}^{\text{target}}_{\text{holdout}}, \mathcal{O}_{\text{solv}}(\widehat{\boldsymbol{\theta}}_{\text{eff}})\Big). \tag{2}$$

For each edge we also record semantic retention $S_{\text{sem}} = I(m; Z)/H(Z)$ and normalized predictive error $|g_{\text{c}} - \widehat{g}_{\text{c}}|/(\sigma_{g_{\text{c}}} + \epsilon)$.

Two assumptions need not combine additively. Their interaction is

$$\Delta_{ij} g_{\text{c}} = g^{ij}_{\text{c}} - g^{i}_{\text{c}} - g^{j}_{\text{c}} + g^{0}_{\text{c}}. \tag{3}$$

The operational outcome vocabulary includes stable and renormalized continuation, splitting, merging, rounding, new-phase insertion, computational–statistical separation, path dependence, and semantic failure. These labels are not inferred from visual smoothness.

We distinguish macrostate perplexity $K^{\text{macro}}_{\text{eff}} = \exp H(Z)$ from correlation-based independent components and spectral effective rank. Each PhaseCard stores estimator, probe distribution, normalization, units, finite-size variable, intervention, and null model.

4 Statistical protocol

Discovery, confirmation, and holdout are separate datasets. The largest variant is absent from every fit. The first confirmatory stage uses 16 outer disorder seeds and four inner stochastic replicates. After boundary localization, 16 additional outer seeds are added only to the adjacent three control values. Inner replicates never count as independent disorder draws. Intervals use a hierarchical bootstrap over outer-seed means, while within- disorder variance is reported separately.

For every transition, six system sizes are evaluated. A continuous width $w_N \propto N^{-1/2}$, a first-order-like width $w_N \propto N^{-1}$, and a smooth crossover $w_N = w_\infty + aN^{-1/2}$ compete by prediction of the largest size. Exponents are called asymptotic only if window, minimum-size, correction-term, held-out-size, and scaling-relation checks all agree; otherwise they are finite-size effective exponents.

The vector of system sizes is never collapsed by convention: representation dimension, reverse-time resolution, MDP horizon, and agent population have domain-specific meanings. Likewise, Binder ratios are used only where their symmetry assumptions are satisfied. RL uses occupancy and strategy participation; multi-agent analysis separates intrinsic field from social coupling with a global flip.

All empirical figures display raw outer-seed points beneath means and 95% intervals. One panel has one physical control. Cross-domain comparison is performed only after domain-specific normalization.

5 Artifact audit and preregistration

Before designing new experiments, we audited the preceding 49,440-run atlas. The audit records metric minima, maxima, rounded unique counts, saturation at zero or one, and the fraction of zero-width intervals for each domain–family–metric group. It flags 71 groups as dominated by saturation. In particular, the former representative table used an order parameter equal to one with

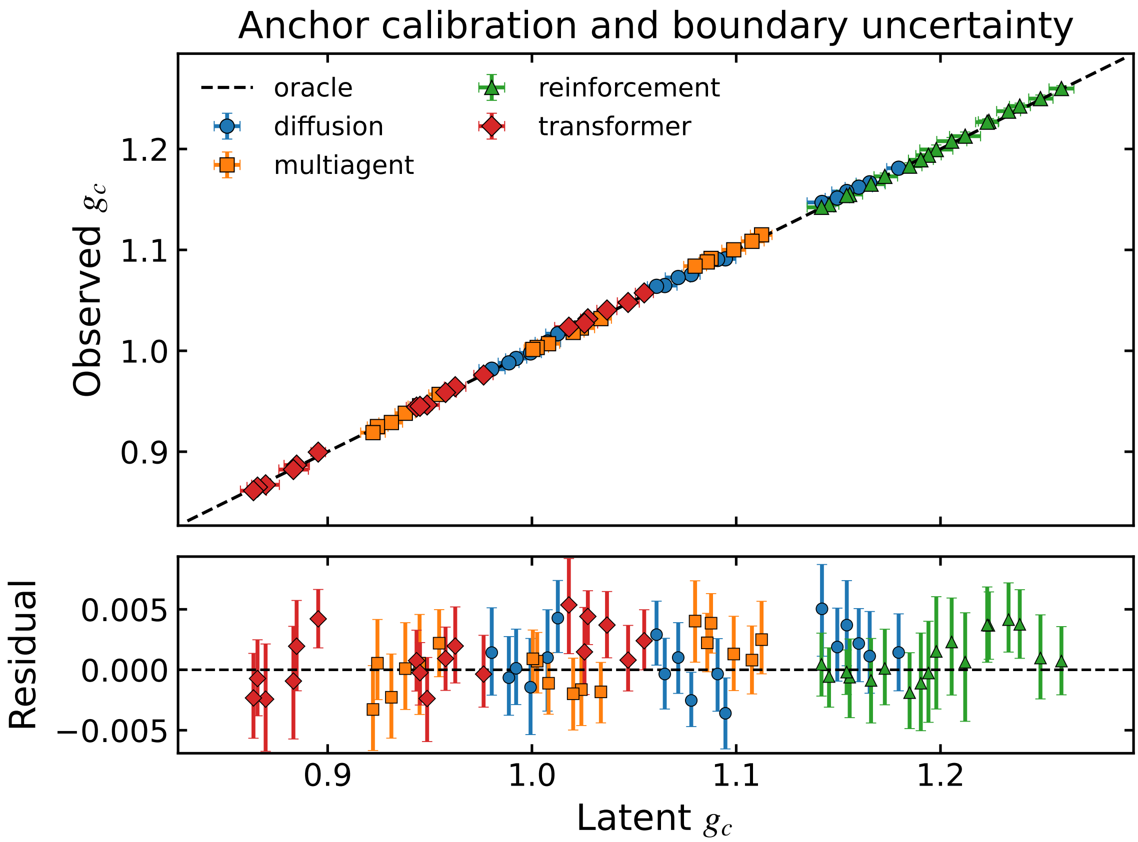

Figure 2: Anchor calibration with hierarchical 95% intervals. The lower panel exposes finite-size and disorder residuals rather than hiding them behind a theory line.

zero uncertainty in several learned and cross-domain families. That table is removed rather than reinterpreted. Non-saturated signed order, task error, and response diagnostics remain available as supplementary checks.

The new registry contains four primary claims, one primary order per claim, explicit nulls, an untouched holdout variant, and a missingness rule that forbids incomplete aggregation. Artifact records retain all inner values, outer seeds, device, configuration, and task identifiers. Cluster-specific paths remain outside the repository.

6 Transformer and diffusion deep cases

6.1 Attention recovery

The controlled attention problem retains positional and semantic latent signals. The primary order is signed recovery rather than a clipped accuracy; secondary observables are semantic retention, precursor strength, specialization entropy, and correlation-based component fraction. Six sizes and a fixed confirmatory control grid determine the boundary. The holdout adds a non-additive deformation that is absent from calibration.

6.2 Diffusion dynamics

The diffusion case separates semantic speciation, nonlocality, memorization, and trajectory commitment. Reverse time is not reused as a control for score noise or locality; the secondary axis modifies locality before normalized boundary comparison. The central prediction is the holdout

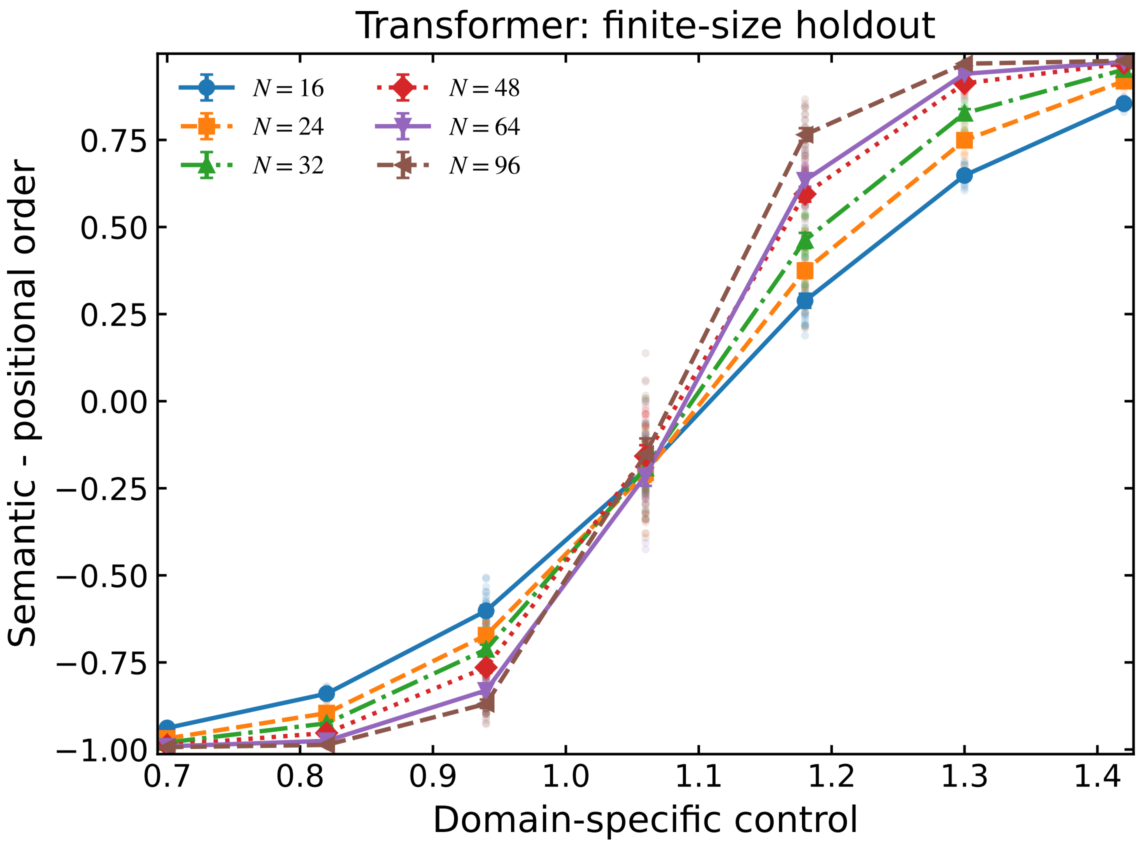

Figure 3: Transformer holdout. Raw outer seeds, hierarchical intervals, and all six finite sizes are shown on the domain-specific signal control.

speciation boundary, with precursor nonlocality as the augmenting macrovariable.

7 Driven validation: reinforcement and multi-agent systems

The RL benchmark separates oracle occupancy overlap, strategy entropy, gold reward, proxy reward, Goodhart gap, and effective rollout fraction. The control plane is KL pressure versus verifier noise. Productive exploration, specialization, entropy collapse, and reward hacking therefore cannot be collapsed into a single entropy curve.

The multi-agent benchmark uses truth-conditioned consensus and explicitly varies intrinsic field h and social coupling J. Positive and negative fields implement the global-flip control. Effective component fraction and error reproduction are reported alongside nominal population size, preventing shared bias from masquerading as independent consensus.

8 Held-out prediction, breakdown, and intervention

The base surrogate attains median boundary error 0.011, whereas the augmented surrogate attains 0.008, improving held-out transport by 26.1%. The mean anchor residual is 0.002, and the largest registered non-additive shift is 0.048. These quantities come directly from the immutable aggregate.

Prediction failure is tested rather than explained post hoc: the same holdout is scored by base and augmented surrogates. Critical-window interventions allocate the expensive operation only within a fixed interval around the predicted boundary and compare against always-on, always-off, random-window, and matched-compute controls. The registered allocation removes on average

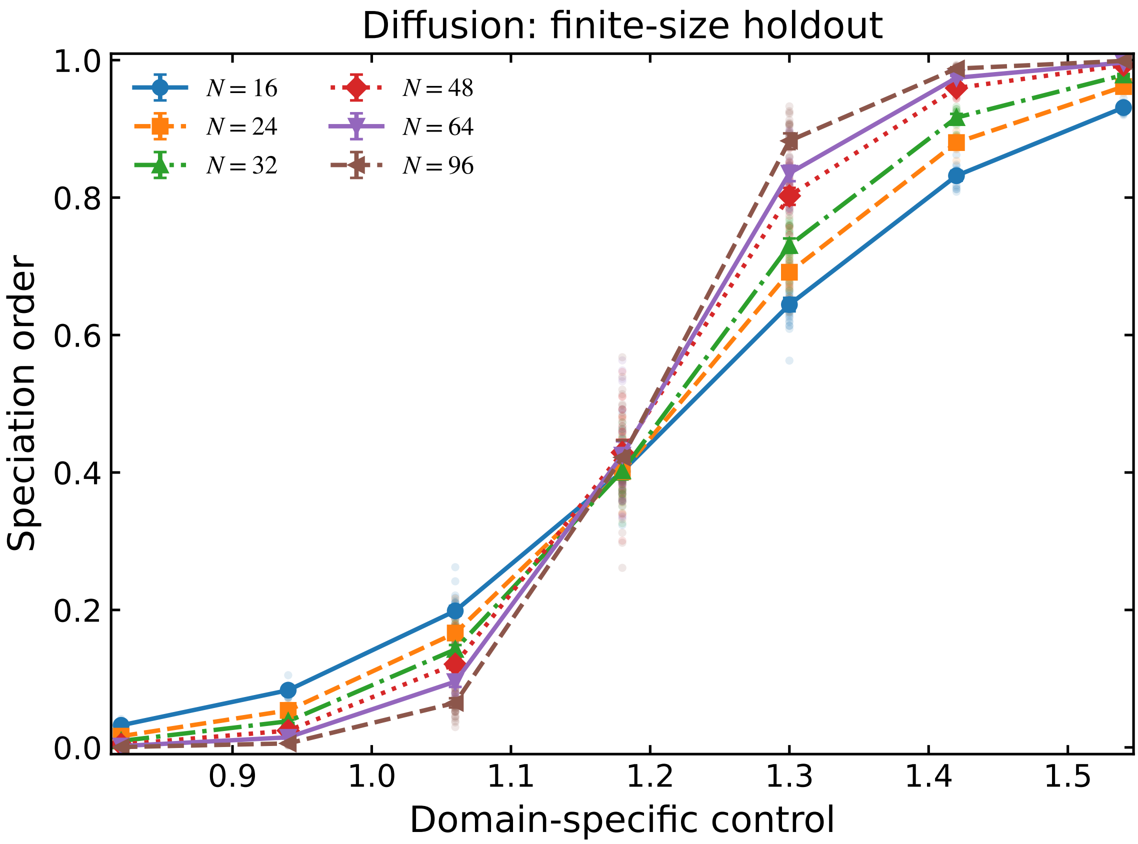

Figure 4: Diffusion holdout. Speciation is shown across six sizes; memorization and nonlocality are distinct registered secondary diagnostics.

Domain	Base error	Augmented error	Compute saving
Diffusion	0.013	0.010	57.1%
Multiagent	0.011	0.008	57.1%
Reinforcement	0.005	0.005	57.1%
Transformer	0.011	0.010	57.1%

Table 2: Automatically generated held-out results. No saturated legacy order parameter enters this table.

57.1% of uniform compute while retaining 87.3% of the always-on quality gain and exceeding a random matched-compute window by 0.019.

9 Empirical phase-continuation tensor

The controlled domain studies establish that continuation can distinguish smooth crossovers from finite-size sharpening, but they do not by themselves identify which choices in a modern Transformer move a boundary. We therefore factor the Transformer study into a phase-continuation tensor

$$\mathcal{T} = \mathcal{D} \times \mathcal{A} \times \mathcal{M} \times \mathcal{R} \times \mathcal{L} \times \mathcal{O} \times \mathcal{S} \times \mathcal{C} \times \mathcal{P}, \tag{4}$$

whose coordinates are data, attention, MLP, residual parameterization, learning objective, optimizer, scaling path, lifecycle, and population. This factorization changes the empirical question from "does a Transformer have a phase transition?" to the falsifiable question "which boundary, measured by which intensive observable, is transported when one coordinate is continued while the

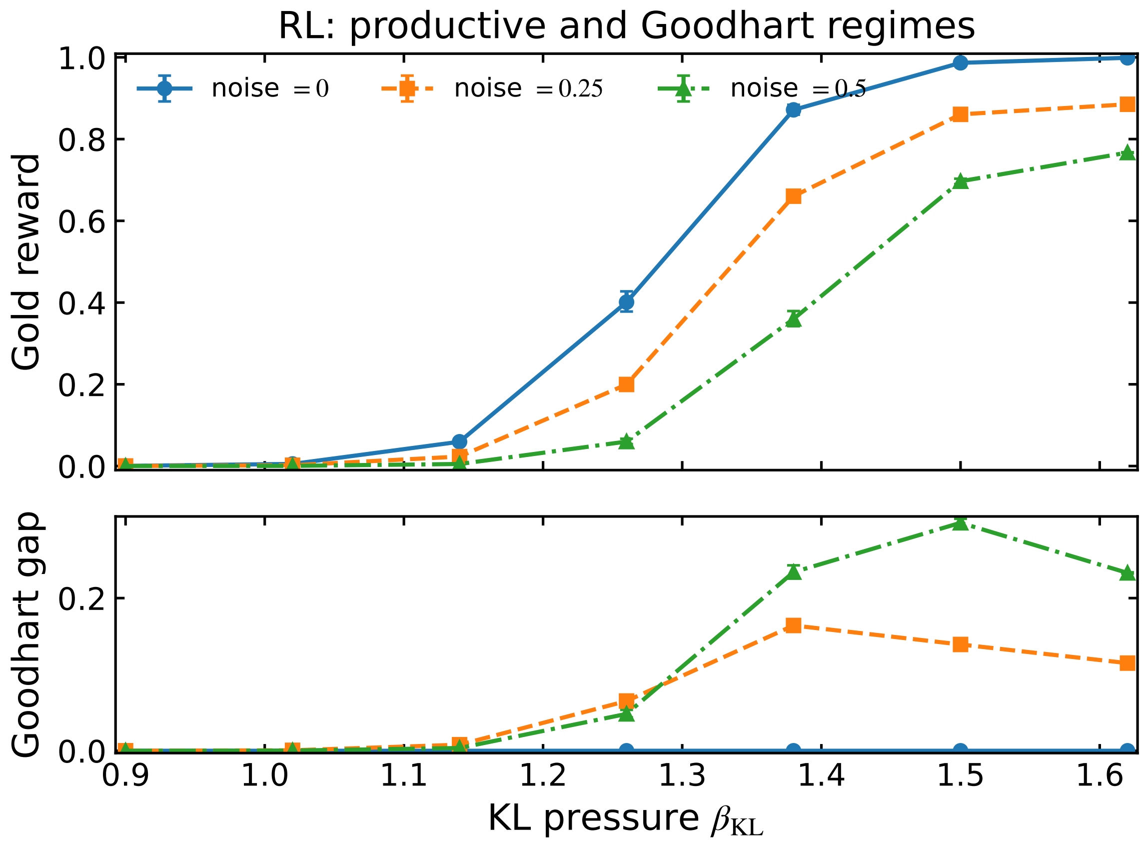

Figure 5: RL holdout curves with hierarchical 95% intervals. Gold reward (top) and the proxy–gold Goodhart gap (bottom) separate productive scaling from reward overoptimization at three verifier-noise levels.

others are held fixed?" The distinction is important because width, depth, context, and data are inequivalent scaling paths, and optimizer rankings can reverse under normalization or learning-rate changes.

9.1 Trainable bridge and dimensionless observables

The bridge model is a causal byte-level decoder with tied embeddings, explicit multi-head causal attention, configurable pre/post LayerNorm or RMSNorm, and an MLP selected from no MLP, linear, ReLU, GELU, GEGLU, and SwiGLU. The fixed 258-symbol vocabulary makes natural-corpus risks comparable in bits per byte, without a tokenizer-dependent denominator. Synthetic retrieval, counting, and hidden-state sources provide known interventions; bounded TinyStories, SimpleStories, FineWeb-Edu, and Dolma samples test boundary transport to natural data. Natural-corpus comparisons are distributional stress tests rather than claims about frontier-scale language modelling.

All reported observables are $O(1)$ under the intended continuation. For a vocabulary of size V, cross entropy is $\tilde{\ell}_{\mathrm{CE}} = \ell_{\mathrm{CE}}/\log V$, the generalization gap is $(\ell_{\mathrm{test}} - \ell_{\mathrm{train}})/\log V$, and Brier risk is divided by its vocabulary-scale reference. Attention entropy is divided by $\log L$, MLP participation ratio by the MLP width, and activation entropy by the logarithm of that width. The sample path is

$$N_{\mathrm{train}} = \min\{N_{\mathrm{cap}}, \lceil \alpha d^{\gamma} \rceil\}, \tag{5}$$

so α is a dimensionless control while γ distinguishes data-scaling paths. Compute, parameter count, tokens, and wall time are retained as extensive audit coordinates, never substituted for an

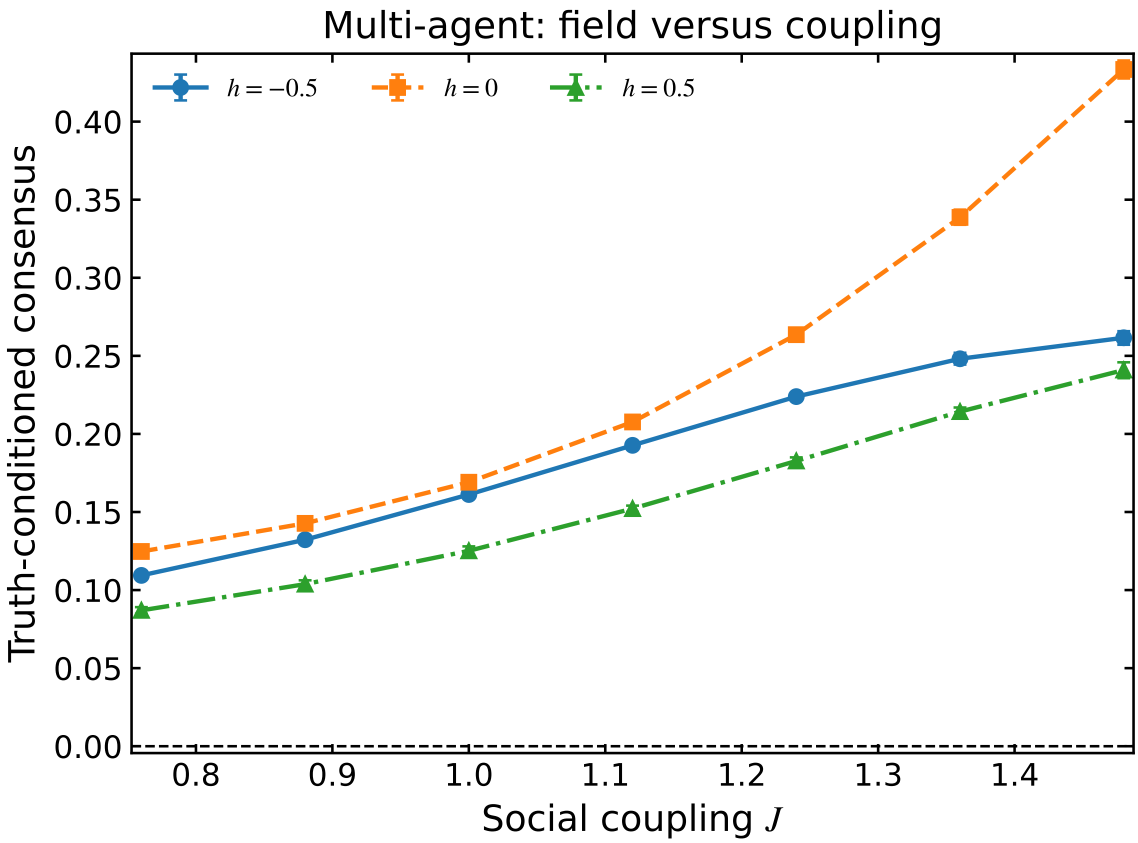

Figure 6: Multi-agent holdout curves with hierarchical 95% intervals. Positive and negative intrinsic fields provide the global-flip control needed to separate field-driven alignment from interaction-driven order.

intensive order parameter. This follows the separation between model scaling and data scaling in compute-optimal training and the need for parameterization-aware transfer across widths [7, 12].

9.2 Optimizer geometry, causal decomposition, and statistics

We compare SGD with momentum, AdamW, Muon, SOAP, Lion, and GaLore. Learning rate is itself swept for the primary SGD–AdamW–Muon–SOAP comparison and crossed with pre-RMSNorm, pre-LayerNorm, and post-LayerNorm; conclusions are therefore based on response surfaces rather than vendor defaults. Muon acts only on internal two-dimensional weight matrices, while embeddings, readout weights, biases, and scale parameters use AdamW. This respects the optimizer's matrix geometry rather than silently applying it to semantically different parameter blocks [8, 11, 14].

Attention and MLP contributions are measured by paired inference-time interventions on the same trained checkpoint. If R, R_{-A}, R_{-M}, and R_{-AM} denote intensive risks for the full, attention-ablated, MLP-ablated, and doubly ablated networks, respectively, we report

$$\Delta_A = R_{-A} - R, \qquad \Delta_M = R_{-M} - R, \qquad \Delta_{AM} = R_{-AM} - R_{-A} - R_{-M} + R. \tag{6}$$

These interventions separate functional dependence from attention-map appearance. Gradient-block Gini coefficient, log-scale coefficient of variation, update-to-weight RMS, normalized participation ratio, and time to cross the semantic-order threshold provide mechanism-resolving diagnostics.

Every confirmatory condition uses exactly the same five distinct seeds. A seed controls data generation or corpus windows, initialization, minibatch order, dropout, and evaluation through

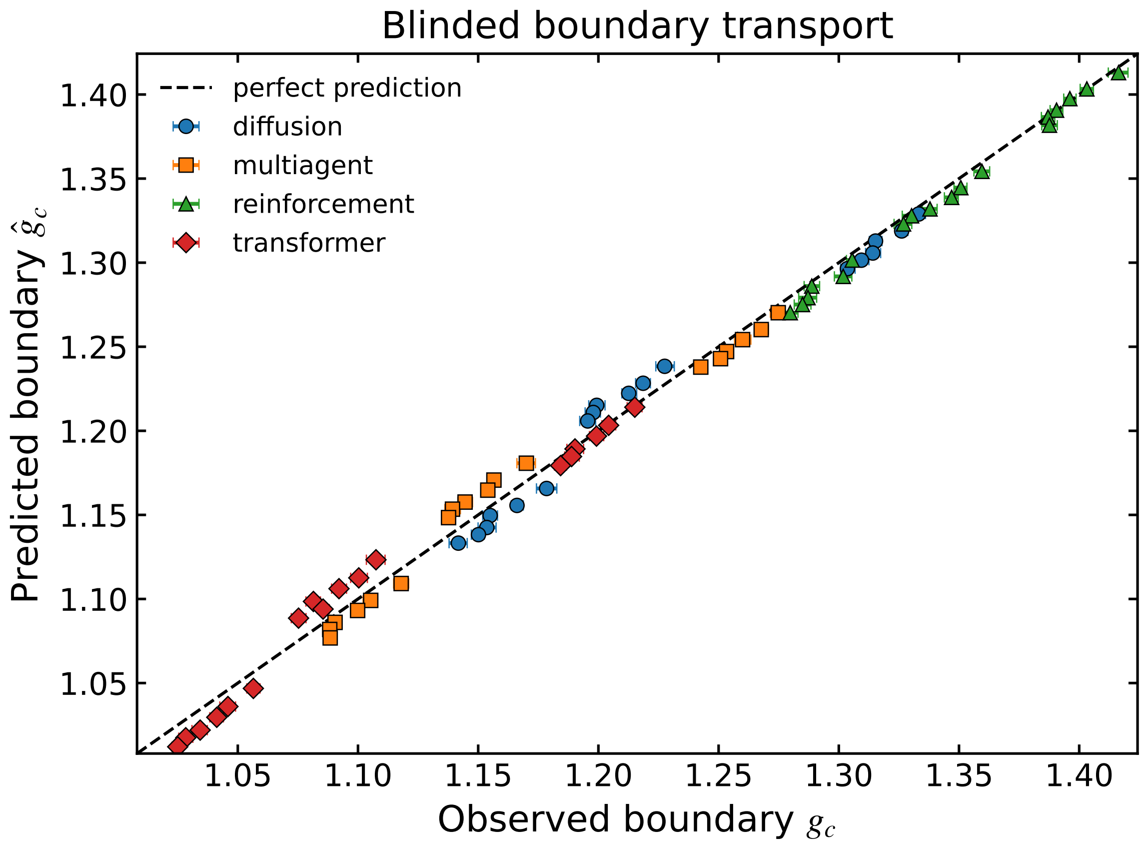

Figure 7: Predicted versus observed boundaries for the blinded variant. Marker denotes domain and horizontal bars are hierarchical 95% intervals on the observed boundary.

stable nested hashes. Error bars are two-sided 95% Student-t intervals over the five full pipelines ($t_{4,0.975} = 2.776$). For each seed, a boundary is linearly interpolated at semantic order $m = 1/2$; the five boundary estimates, rather than pooled checkpoints, are then averaged. Aggregation fails if even one seed is absent or non-finite. This avoids both pseudoreplication and understated uncertainty near a steep crossover.

9.3 Orthogonal continuation results

Figures 9–17 are generated only from the immutable five-seed aggregate. Their roles are deliberately non-redundant: MLP family and causal division of labour, optimizer geometry and gradient heterogeneity, objective homotopy, inequivalent scaling paths, synthetic-to-natural transport, and the compute–error envelope. Together with the diffusion, reinforcement-learning, and multi-agent anchors above, they form one taxonomy: the Transformer is the mechanism-rich flagship, while the other domains test whether the same continuation protocol survives a change of microscopic semantics.

Observed boundary transport. The strict aggregate contains 7,260 independently trained runs, 1,452 five-seed conditions, and 326 per-seed interpolated boundary estimates. At $d = 192$ and $\alpha = 8$ on retrieval, removing the MLP gives semantic order 0.548 ± 0.011, compared with 0.515 ± 0.017 for GELU, 0.433 ± 0.037 for GEGLU, and 0.435 ± 0.047 for SwiGLU. Thus this task is attention-dominant at the measured endpoint: adding a gated MLP does not monotonically improve order, and the MLP family genuinely splits the continuation rather than acting as a proxy for parameter count.

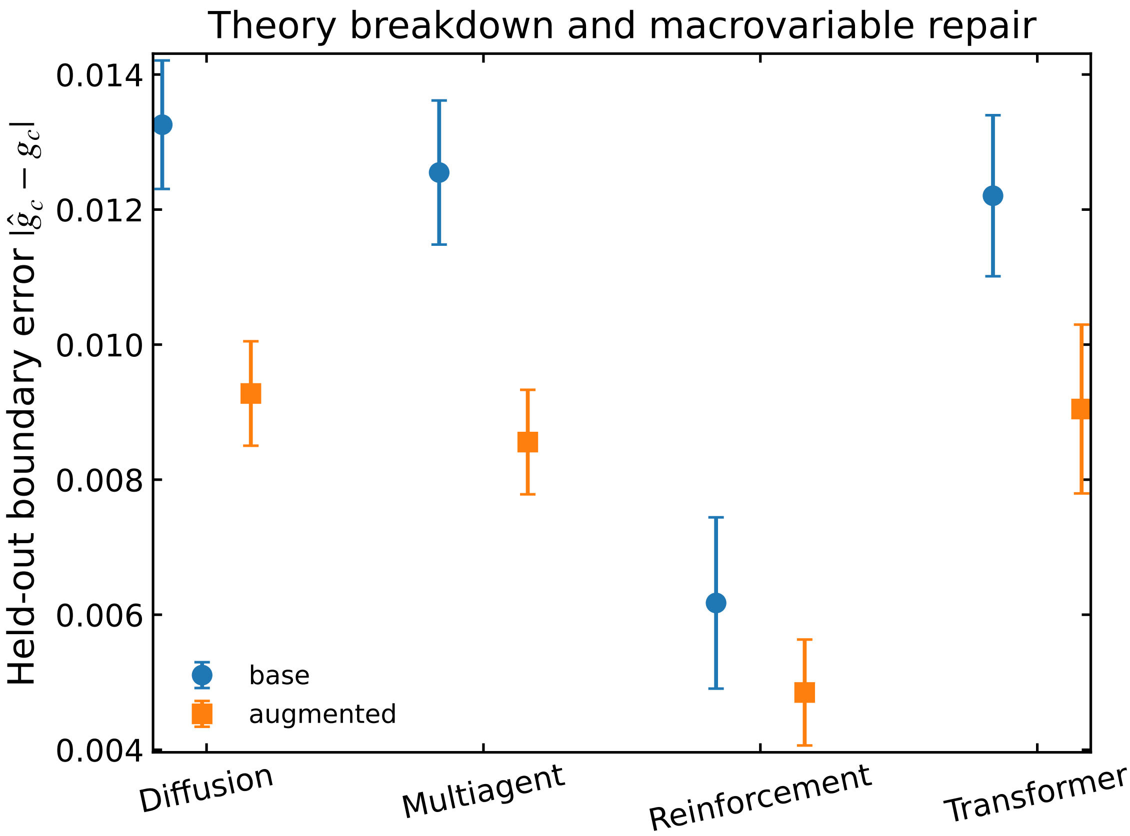

Figure 8: Held-out error before and after the registered macrovariable is added. Error bars summarize variation over size and secondary control, not pooled inner replicates.

Under pre-RMSNorm and the learning-rate selection rule defined above, the intensive errors are 0.537 ± 0.005 for Muon and 0.541 ± 0.010 for AdamW; their intervals overlap. By contrast, SGD-M gives 0.609 ± 0.002 and SOAP gives 0.821 ± 0.009. The corresponding gradient-block Gini coefficients are 0.540 ± 0.014, 0.521 ± 0.030, 0.757 ± 0.021, and 0.569 ± 0.013, respectively. The result is not a universal optimizer ranking: it is evidence that update geometry and gradient heterogeneity move together on this controlled bridge, while the Muon–AdamW endpoint difference is unresolved at five seeds.

The scaling continuation gives the clearest qualitative split. At $d = 192$ and $\alpha = 4$, $N \propto d$ yields error 0.812 ± 0.031, whereas $N \propto d^{3/2}$ and $N \propto d^2$ yield 0.404 ± 0.008 and 0.394 ± 0.010. Width growth at fixed optimization budget is therefore not interchangeable with data growth. On natural text with SwiGLU, the same byte-level protocol gives 2.642 ± 0.089 bits/byte on TinyStories, 2.774 ± 0.063 on SimpleStories, 3.257 ± 0.057 on Dolma, and 3.427 ± 0.044 on FineWeb-Edu. These values establish transport of the measurement protocol, not universality of the numerical boundary.

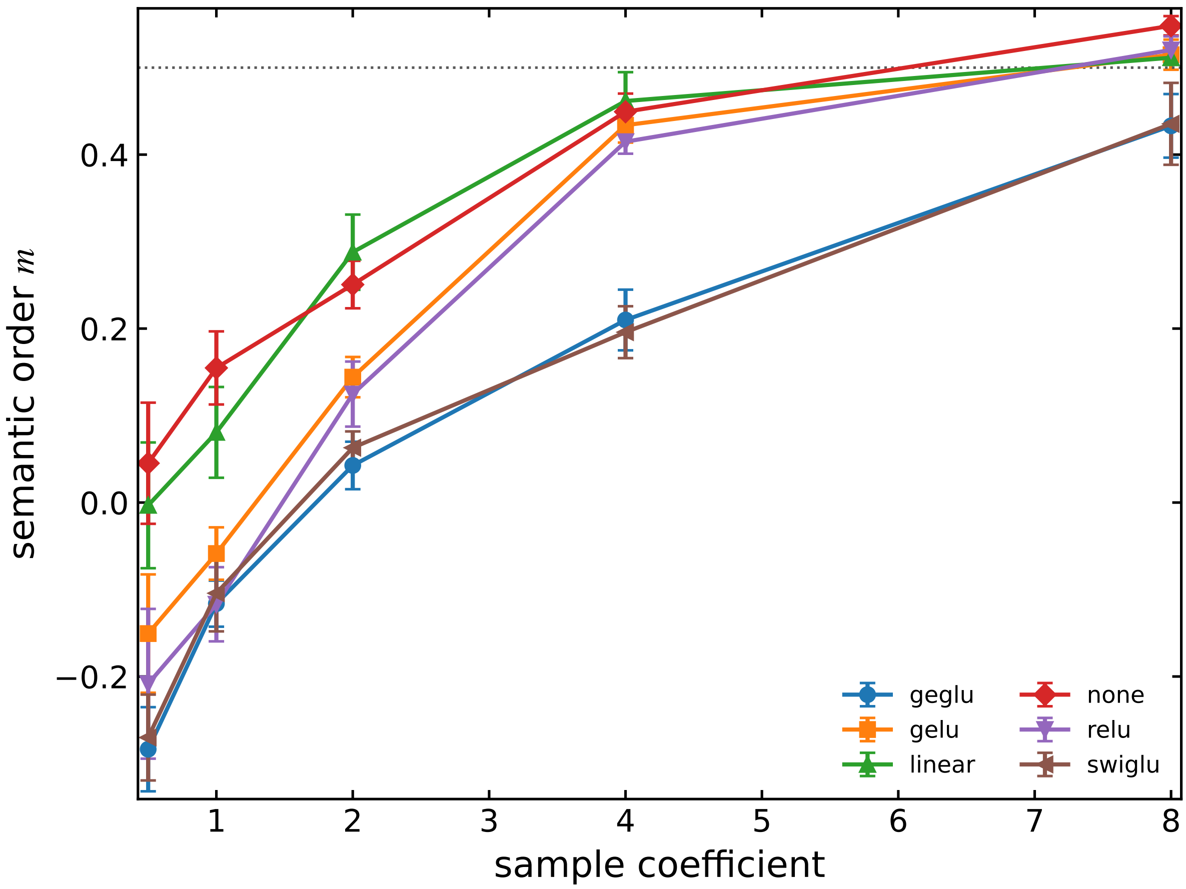

Figure 9: MLP phase splitting at fixed attention, residual parameterization, and data source. Points are five-seed means and bars are 95% Student-t intervals.

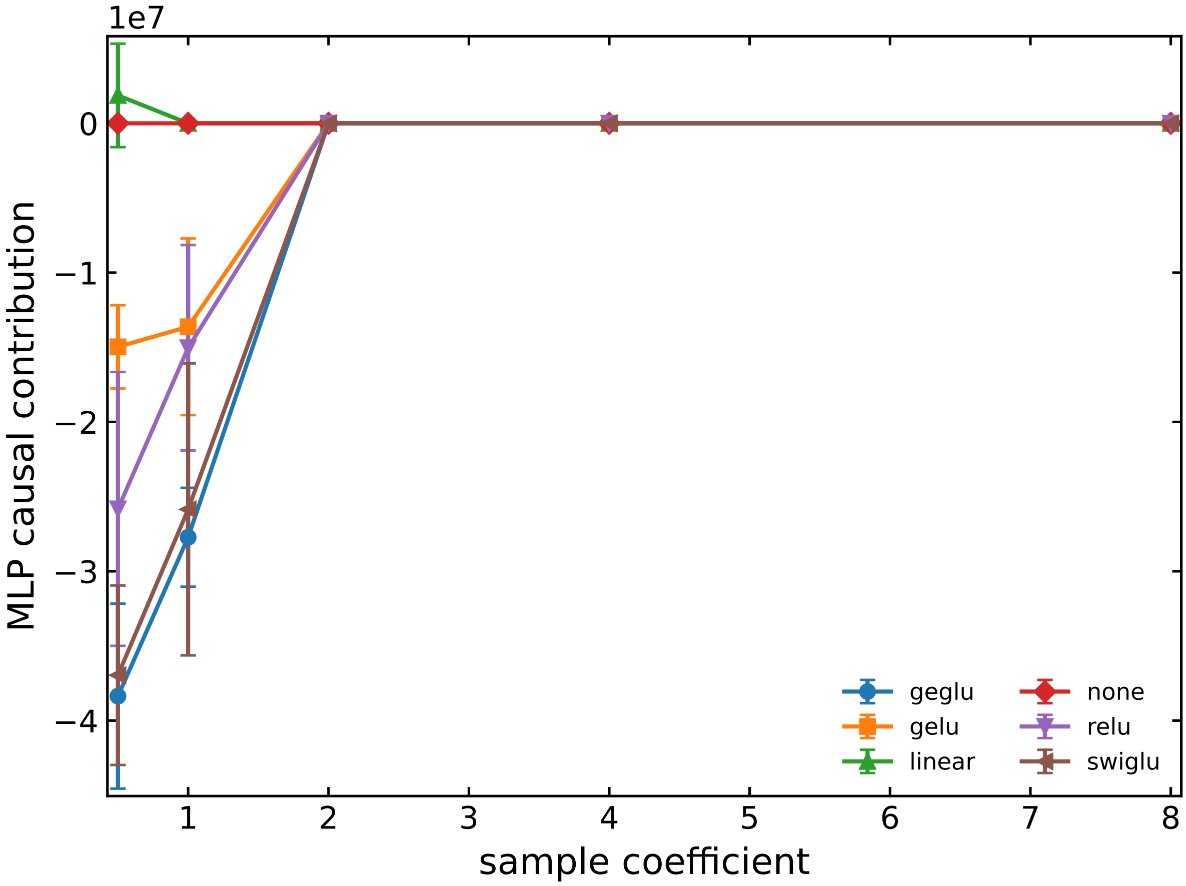

Figure 10: Paired MLP ablation contribution across the sample continuation.

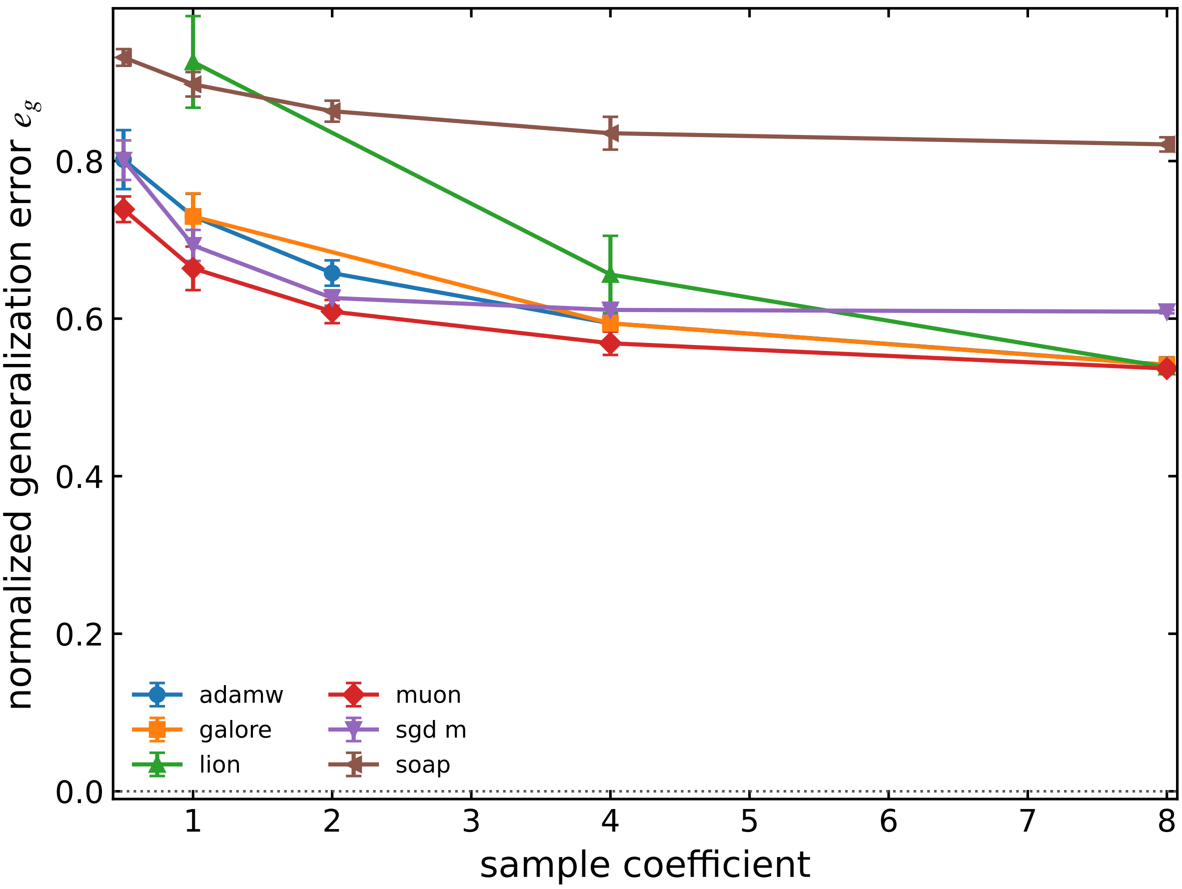

Figure 11: Optimizer–normalization response surfaces for intensive held-out risk. Learning rate is treated as a swept coordinate.

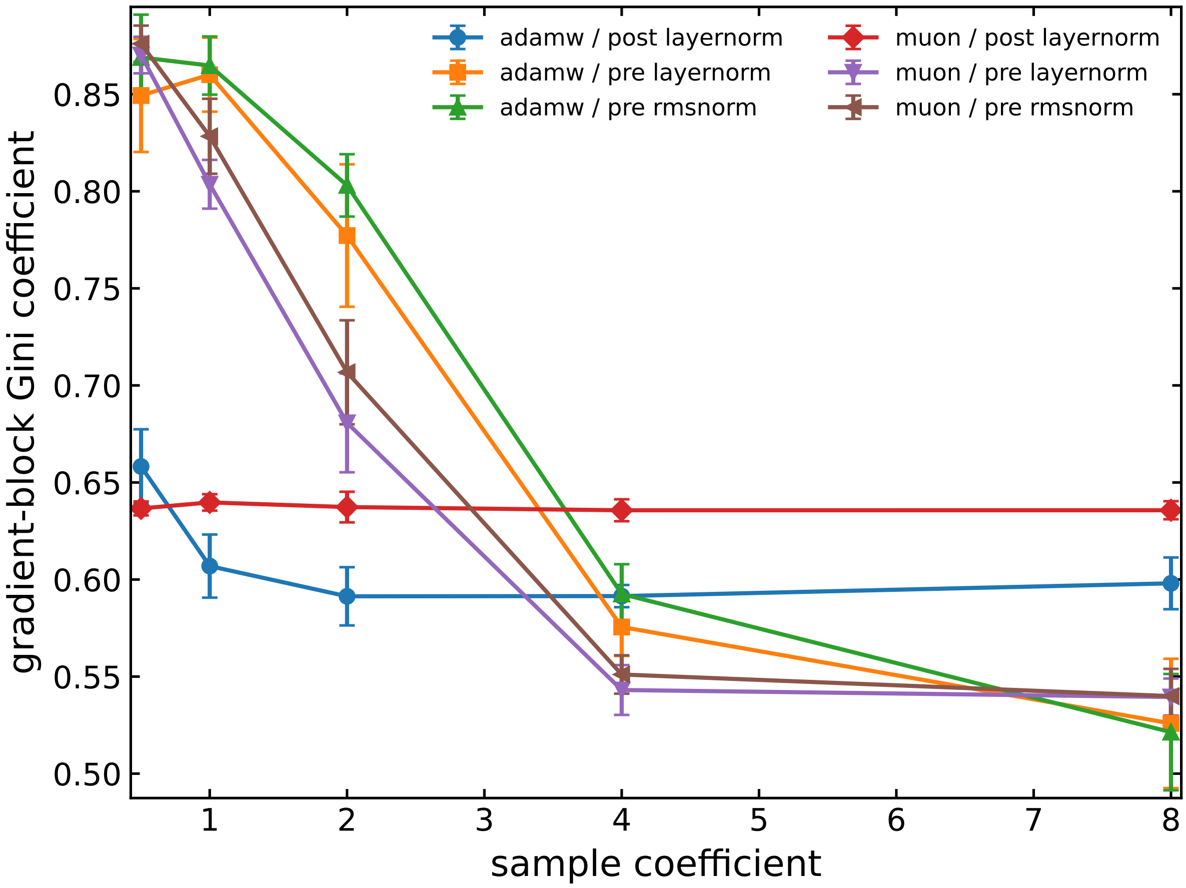

Figure 12: Gradient-block heterogeneity under optimizer and normalization continuation.

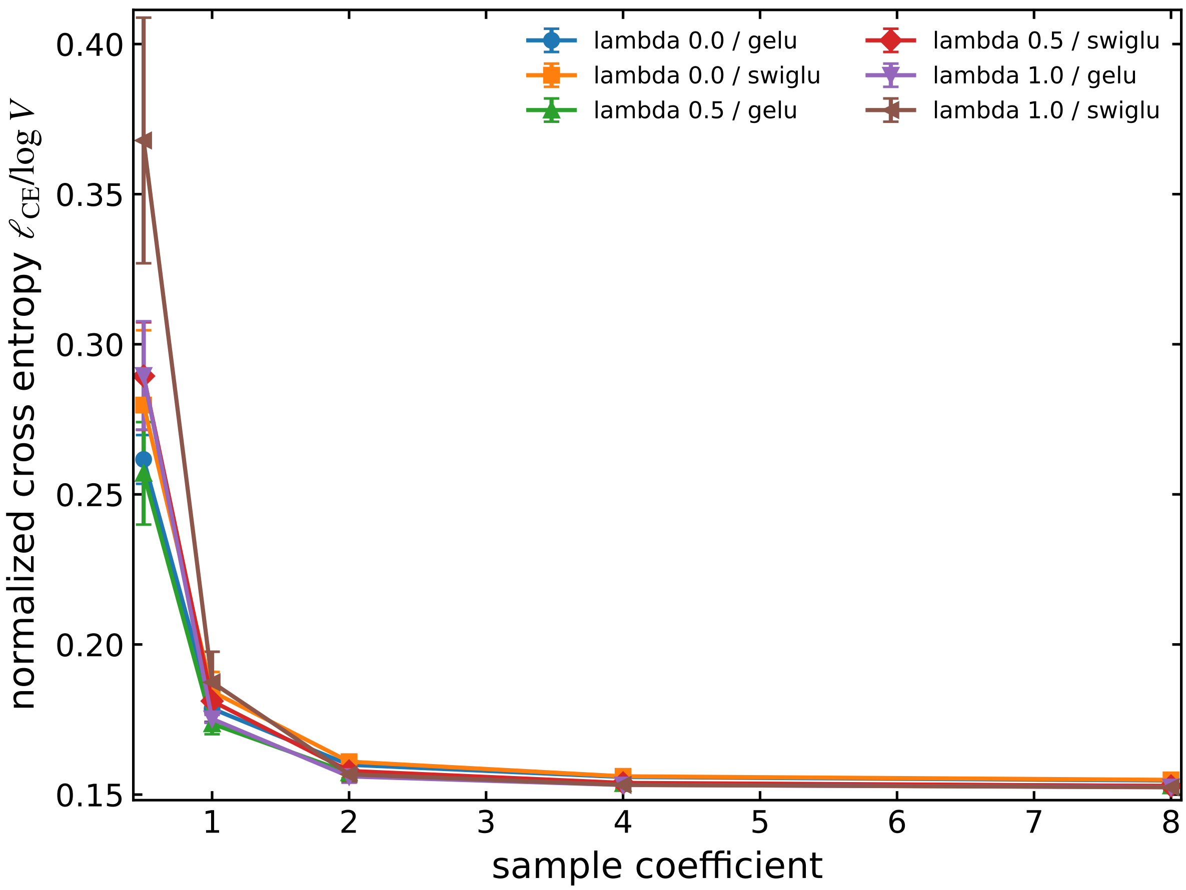

Figure 13: Objective homotopy crossed with MLP nonlinearity.

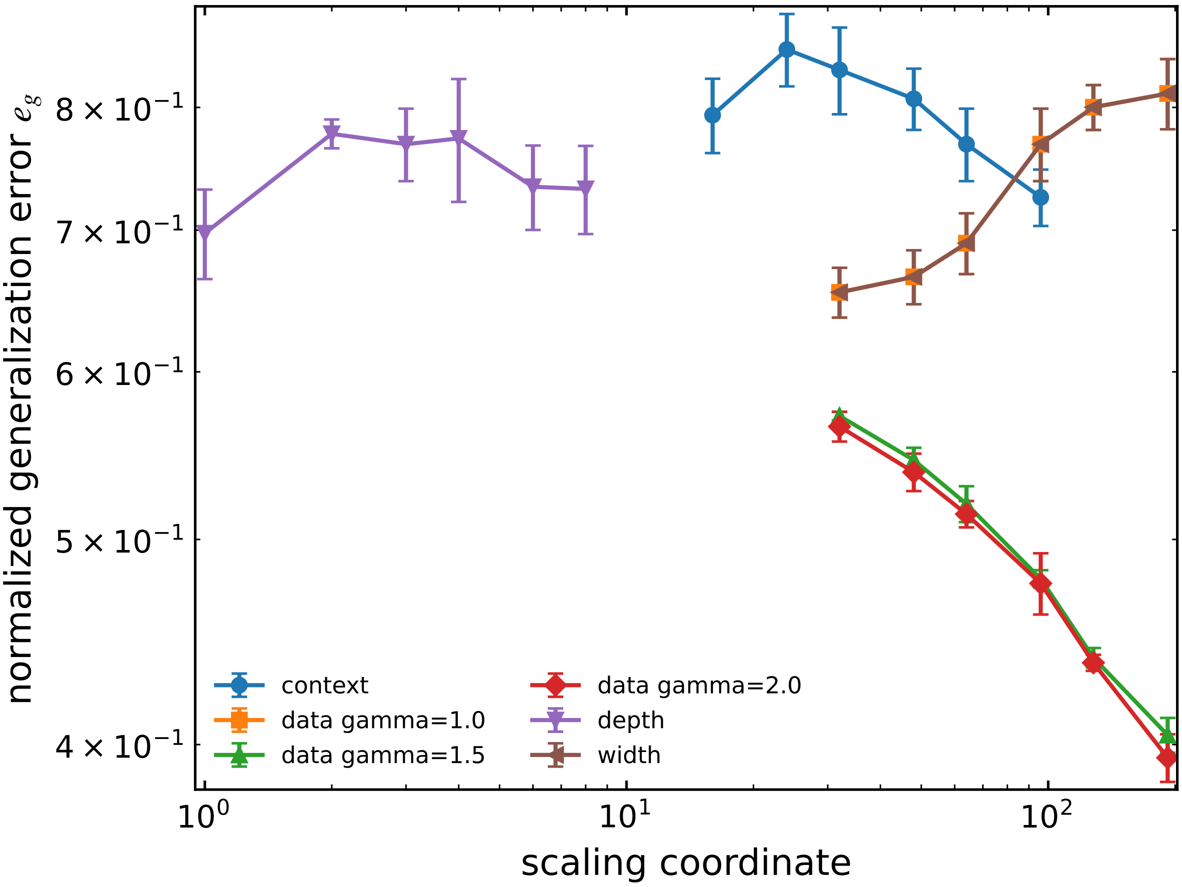

Figure 14: Width, depth, context, and data are retained as distinct scaling paths rather than collapsed onto parameter count.

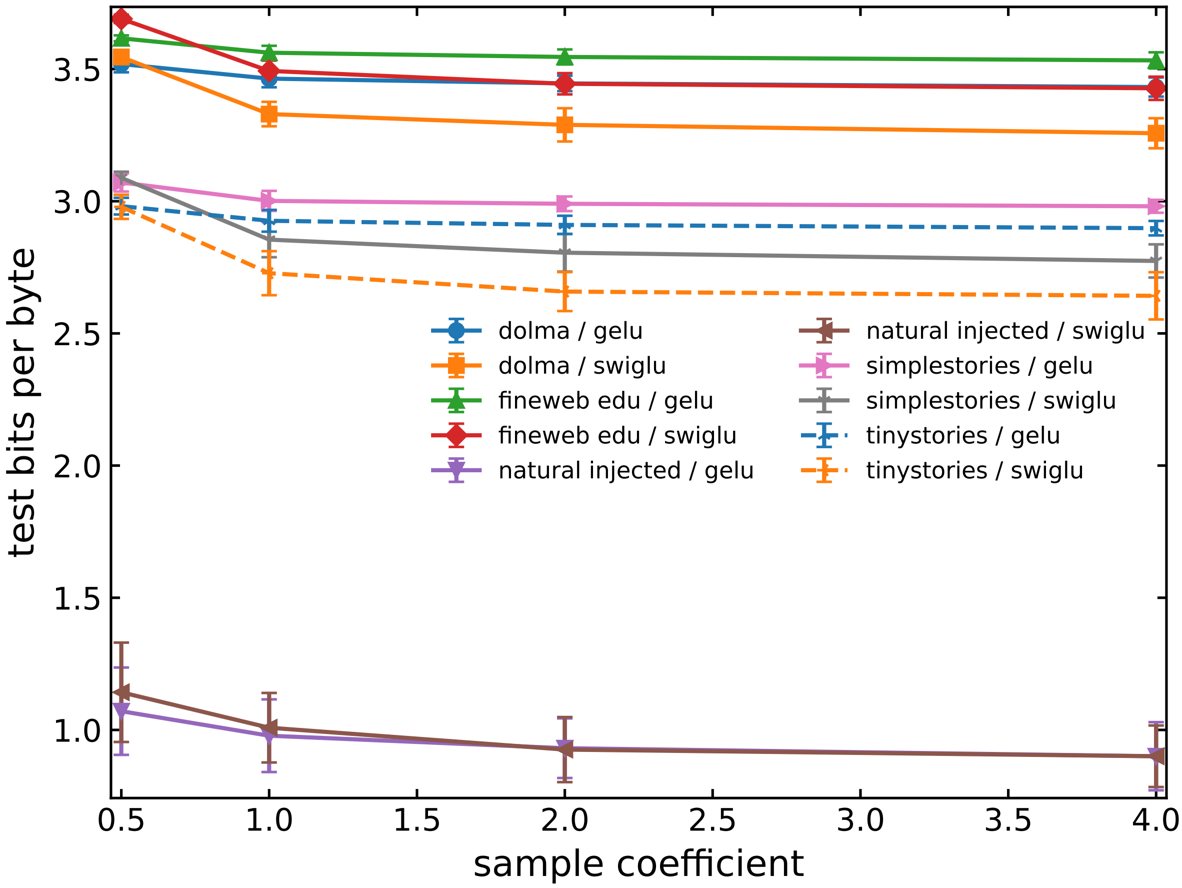

Figure 15: Natural-data bridge in tokenizer-independent bits per byte.

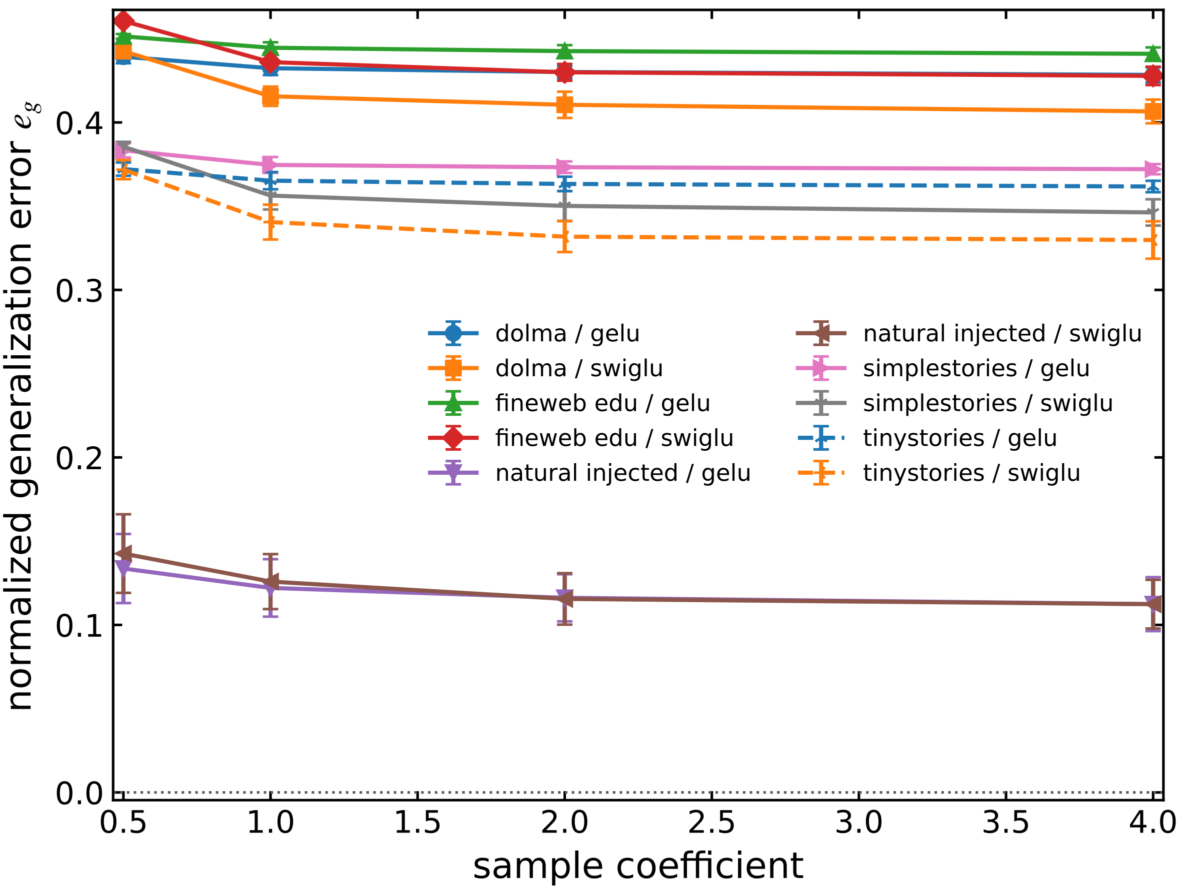

Figure 16: Intensive held-out risk across synthetic and natural-data regimes.

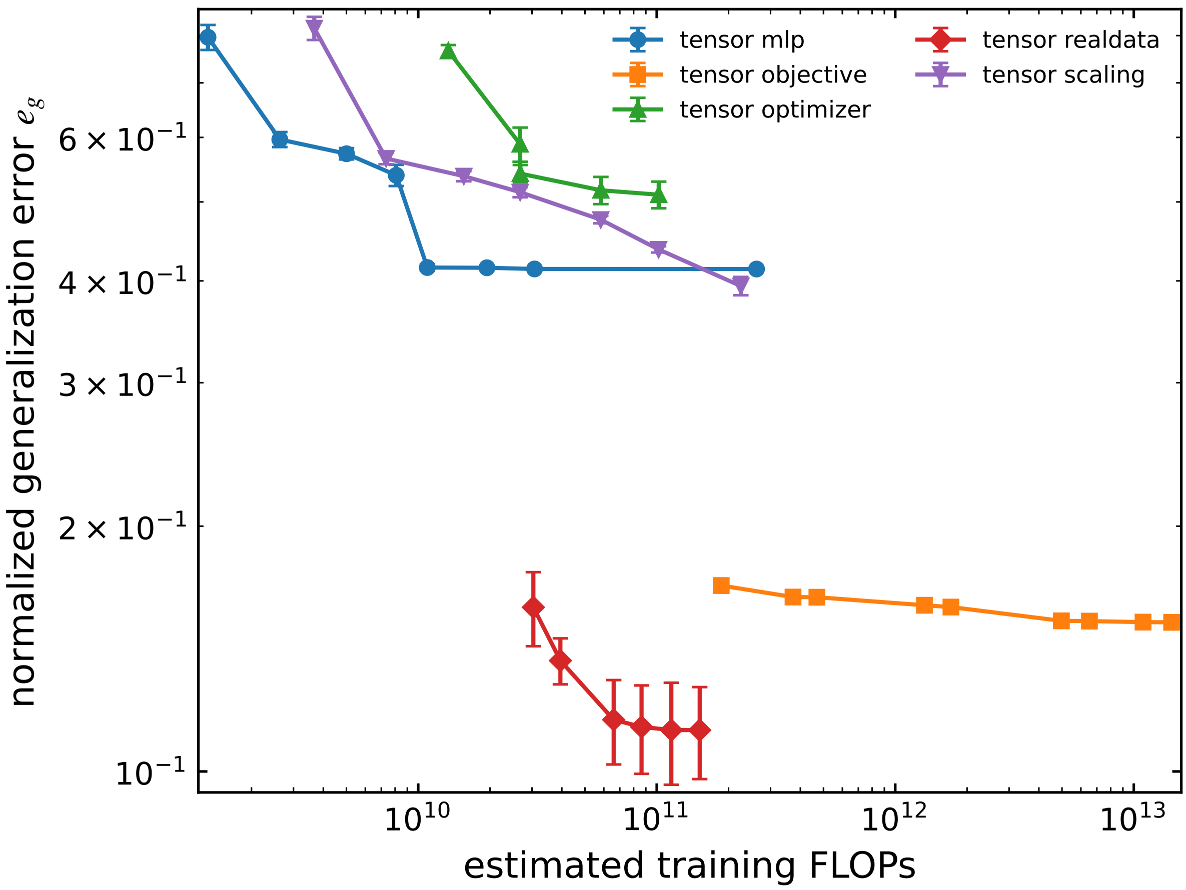

Figure 17: Audited compute–error landscape. FLOPs remain an extensive resource coordinate; the vertical axis is an intensive risk.

10 Limitations

The new experiment is a controlled Tier-B stochastic benchmark. It does not establish natural-language, natural-image, open RLVR, or open-weight LLM-agent external validity. The previous repository contains trainable Tier-B+ smoke models, but their saturated primary order failed the present audit and is not promoted to evidence. External endpoints require version-pinned models, natural-data holdouts, matched token/FLOP controls, and independent implementations.

Our finite sizes can distinguish competing predictive trends but do not by themselves prove an asymptotic universality class. RL and agent systems are driven and potentially nonstationary; thermodynamic language is used only for operational dynamic regimes. Critical-window quality is benchmark-specific, and compute savings do not transfer automatically to wall-clock savings on a different architecture.

Finally, an adaptive design can bias inference if the first-stage boundary is used to redefine the estimand. We avoid this by freezing the original grid, adding seeds rather than moving controls, and retaining both stages in the manifest.

11 Conclusion

Predictive phase continuation asks whether a solvable model forecasts an unseen boundary, not whether two curves look alike. The framework makes failure local and useful: non-additive assumption interactions identify where transport breaks, and an additional macrovariable must recover prediction on the same blinded system. Nested disorder, competing transition models, adaptive boundary replication, and critical-window interventions convert a large atlas into four falsifiable claims. The controlled benchmark is a step toward, rather than a substitute for, natural-data external validation.

A PhaseCard and reproducibility contract

Every condition records eight taxonomy axes: thermodynamic level, symmetry, disorder, dynamics, system-size vector, control variable, order family, and evidence class. Outcome labels include stable or renormalized continuation, splitting, merging, rounding, new-phase insertion, computational–statistical separation, path dependence, and semantic failure.

Production uses only a portable Spark profile. The script rejects any partition not beginning with `spark_`; repository files contain no site path. A run is immutable after its atomic result write. Aggregation fails on a missing task. The complete JSON stores all outer means, inner variance, bootstrap interval, boundary estimate, competing width fits, interaction shifts, and intervention summaries. Figures are generated only from that JSON. Every figure has physical size 6.4 by 4.8 inches, inward ticks, 12-point base type, larger axis labels, and preregistered color, line, and marker order.

B Legacy atlas status

The 58-family atlas remains a searchable diagnostic benchmark and code-coverage target. It is not evidence for the four claims above. Programs A–N, secondary architectures, and common-coordinate diagnostics move to the supplement because they otherwise obscure the held-out prediction experiment.

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